Homework I — Solutions

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Contents

Problem 1 **2**
1.A) 2
1.B) 2

Problem 2 **3**
2.A) 3
2.B) 3

Problem 3 **5**

Problem 4 **7**
4.A) 7
4.B) 7
4.C) 7

Problem 5 **9**

A Conventions **10**
A.A) Metric 10
A.B) Levi-Civita 10

B Lorentz Group Digression **11**

C Determinants **15**

D Noether's Theorem **16**

E Noether's Second Theorem **19**

F Boundary Conditions **21**

G Algebra of conserved quantities **24**

Problem 1

1.A)

Let an element of the Lorentz group be represented by,

$$U(\Lambda) = \exp\left(\frac{\mathrm{i}}{2}\omega_{\mu\nu}M^{\mu\nu}\right) \tag{1.1}$$

this is indirectly defined through the defining representation of the Lorentz group,

$$\Lambda^{\alpha}{}_{\beta}(\omega) \to -\mathrm{i}\frac{\mathrm{d}\Lambda^{\alpha}{}_{\beta}}{\mathrm{d}\omega_{\mu\nu}}\bigg|_{\omega=0} = -\frac{\mathrm{i}}{2}\left(g^{\alpha\mu}g_{\beta}{}^{\nu} - g^{\alpha\nu}g_{\beta}{}^{\mu}\right) \tag{1.2}$$

this fixes the composition law of two transformations, then,

$$U(\Lambda_1)U(\Lambda_2)U^{-1}(\Lambda_1) = U(\Lambda_1\Lambda_2\Lambda_1^{\mathrm{T}}) \tag{1.3}$$

$$-\mathrm{i}U(\Lambda_1)\frac{\mathrm{d}U(\Lambda_2)}{\mathrm{d}\omega_{2\mu\nu}}\bigg|_{\omega_2=0}U^{-1}(\Lambda_1) = -\mathrm{i}\frac{\mathrm{d}U(\Lambda_1\Lambda_2\Lambda_1^{\mathrm{T}})}{\mathrm{d}\omega_{2\mu\nu}}\bigg|_{\omega_2=0} \tag{1.4}$$

$$U(\Lambda_1)M^{\mu\nu}U^{-1}(\Lambda_1) = -\mathrm{i}\frac{\mathrm{d}U(\Lambda_1\Lambda_2\Lambda_1^{\mathrm{T}})}{\mathrm{d}\omega_{3\alpha\beta}}\bigg|_{\omega_3=0}\frac{1}{2}\frac{\mathrm{d}\omega_{3\alpha\beta}}{\mathrm{d}\omega_{2\mu\nu}}\bigg|_{\omega_2=0} \tag{1.5}$$

$$U(\Lambda_1)M^{\mu\nu}U^{-1}(\Lambda_1) = M^{\alpha\beta}\frac{1}{2}\frac{\mathrm{d}\omega_{3\alpha\beta}}{\mathrm{d}\omega_{2\mu\nu}}\bigg|_{\omega_2=0} \tag{1.6}$$

ω_3 is implicitly defined by

$$-\mathrm{i}\frac{\mathrm{d}\left(\Lambda_1\Lambda_2\Lambda_1^{\mathrm{T}}\right)^{\alpha}{}_{\beta}}{\mathrm{d}\omega_{3\mu\nu}}\bigg|_{\omega_3=0} = -\frac{\mathrm{i}}{2}\left(g^{\alpha\mu}g_{\beta}{}^{\nu} - g^{\alpha\nu}g_{\beta}{}^{\mu}\right) \tag{1.7}$$

$$-\mathrm{i}\frac{\mathrm{d}\left(\Lambda_1\Lambda_2\Lambda_1^{\mathrm{T}}\right)^{\alpha}{}_{\beta}}{\mathrm{d}\omega_{2\rho\sigma}}\bigg|_{\omega_2=0}\frac{1}{2}\frac{\mathrm{d}\omega_{2\rho\sigma}}{\mathrm{d}\omega_{3\mu\nu}} = -\frac{\mathrm{i}}{2}\left(g^{\alpha\mu}g_{\beta}{}^{\nu} - g^{\alpha\nu}g_{\beta}{}^{\mu}\right) \tag{1.8}$$

1.B)

Problem 2

2.A)

$$0 = \frac{\delta}{\delta A_\alpha(y)} S[A_\mu] \tag{2.1}$$

$$0 = -\frac{1}{4}\frac{\delta}{\delta A_\alpha(y)} \int \mathrm{d}^4x\, F^{\mu\nu} F_{\mu\nu} \tag{2.2}$$

$$0 = -\frac{1}{2} \int \mathrm{d}^4x\, F^{\mu\nu} \frac{\delta}{\delta A_\alpha(y)} F_{\mu\nu} \tag{2.3}$$

$$0 = -\frac{1}{2} \int \mathrm{d}^4x\, F^{\mu\nu} \big(\partial_\mu g_\nu{}^\alpha \delta^{(4)}(x-y) - \partial_\nu g_\mu{}^\alpha \delta^{(4)}(x-y)\big) \tag{2.4}$$

$$0 = \int \mathrm{d}^4x\, \partial_\mu F^{\mu\nu} g_\nu{}^\alpha \delta^{(4)}(x-y) \tag{2.5}$$

$$0 = \partial_\mu F^{\mu\alpha} \tag{2.6}$$

2.B)

By eq. (D.21),

$$J_I^\mu = \frac{\partial \mathcal{L}}{\partial \partial_\mu \phi(x)} \frac{\mathrm{d}\phi'(x)}{\mathrm{d}a^I}\bigg|_{a^I=0} + \mathcal{L}\frac{\mathrm{d}x'^\mu}{\mathrm{d}a^I}\bigg|_{a^I=0} \tag{2.7}$$

In this case the transformation is,

$$\begin{cases} x & \to x'^\mu = x^\mu + a^\mu \\ A_\alpha(x) & \to A'_\alpha(x') = A_\alpha(x) \end{cases} \Rightarrow \frac{\mathrm{d}x'^\mu}{\mathrm{d}a^\nu}\bigg|_{a^\nu=0} = g^\mu{}_\nu, \quad \frac{\mathrm{d}A'_\alpha(x)}{\mathrm{d}a^\nu}\bigg|_{a^\nu=0} = -\partial_\nu A_\alpha(x) \tag{2.8}$$

Thus,

$$J^\mu{}_\nu = \frac{\partial \mathcal{L}}{\partial \partial_\mu A_\alpha(x)} \frac{\mathrm{d}A'_\alpha(x)}{\mathrm{d}a^\nu}\bigg|_{a^\nu=0} + \mathcal{L}\frac{\mathrm{d}x'^\mu}{\mathrm{d}a^\nu}\bigg|_{a^\nu=0} \tag{2.9}$$

$$J^\mu{}_\nu = -\frac{\partial \mathcal{L}}{\partial \partial_\mu A_\alpha(x)} \partial_\nu A_\alpha(x) + \mathcal{L} g^\mu{}_\nu \tag{2.10}$$

$$J^\mu{}_\nu = \frac{1}{2} F^{\rho\sigma} \frac{\partial F_{\rho\sigma}}{\partial \partial_\mu A_\alpha(x)} \partial_\nu A_\alpha(x) - \frac{1}{4} F_{\rho\sigma} F^{\rho\sigma} g^\mu{}_\nu \tag{2.11}$$

$$J^\mu{}_\nu = \frac{1}{2} F^{\rho\sigma} \big(g_\rho{}^\mu g_\sigma{}^\alpha - g_\rho{}^\alpha g_\sigma{}^\mu\big) \partial_\nu A_\alpha(x) - \frac{1}{4} F_{\rho\sigma} F^{\rho\sigma} g^\mu{}_\nu \tag{2.12}$$

$$J^\mu{}_\nu = \frac{1}{2} F^{\rho\sigma} \big(\partial_\nu g_\rho{}^\mu A_\sigma - \partial_\nu A_\rho g_\sigma{}^\mu\big) - \frac{1}{4} F_{\rho\sigma} F^{\rho\sigma} g^\mu{}_\nu \tag{2.13}$$

$$J^\mu{}_\nu = F^{\mu\sigma} \partial_\nu A_\sigma - \frac{1}{4} F_{\rho\sigma} F^{\rho\sigma} g^\mu{}_\nu \tag{2.14}$$

Let's check that this quantity is conserved when the equations of motion are obeyed,

$$\partial_\mu J^\mu{}_\nu = \partial_\mu F^{\mu\sigma} \partial_\nu A_\sigma + F^{\mu\sigma} \partial_\mu \partial_\nu A_\sigma - \frac{1}{2} F^{\rho\sigma} \partial_\mu F_{\rho\sigma} g^\mu{}_\nu \tag{2.15}$$

$$\partial_\mu J^\mu{}_\nu = \partial_\mu F^{\mu\sigma}\partial_\nu A_\sigma + F^{\mu\sigma}\partial_\mu\partial_\nu A_\sigma - F^{\rho\sigma}\partial_\nu\partial_\rho A_\sigma = 0 \tag{2.16}$$

Yes! But, it isn't gauge invariant, neither symmetric. How to obtain a gauge invariant and symmetric energy momentum tensor?

Problem 3

The transformation is,

$$\begin{cases} x & \to x' = \mathrm{e}^a x \\ \phi(x) & \to \phi'(x') = \mathrm{e}^{-d_\phi a}\phi(x) \end{cases} \Rightarrow \left.\frac{\mathrm{d}x'^\mu}{\mathrm{d}a}\right|_{a=0} = x^\mu, \quad \left.\frac{\mathrm{d}\phi'(x)}{\mathrm{d}a}\right|_{a=0} = -d_\phi\phi(x) - \partial_\mu\phi(x)x^\mu \tag{3.1}$$

First let's check that it's a symmetry,

$$S'_a[\phi'] = \int \mathrm{d}^4x' \left[-\frac{1}{2}\partial'_\mu\phi'(x')\partial'^\mu\phi'(x') - \frac{\lambda}{4!}\phi'^4(x')\right] \tag{3.2}$$

$$S'_a[\phi'] = \int \mathrm{d}^4x\, \mathrm{e}^{4a}\left[-\frac{1}{2}\mathrm{e}^{-2a-2d_\phi a}\partial_\mu\phi(x)\partial^\mu\phi(x) - \mathrm{e}^{-4d_\phi a}\frac{\lambda}{4!}\phi^4(x)\right] \tag{3.3}$$

$$S'_a[\phi'] = \int \mathrm{d}^4x \left[-\frac{1}{2}\mathrm{e}^{2a-2d_\phi a}\partial_\mu\phi(x)\partial^\mu\phi(x) - \mathrm{e}^{4a-4d_\phi a}\frac{\lambda}{4!}\phi^4(x)\right] \tag{3.4}$$

For $d_\phi = 1$,

$$S'_a[\phi'] = \int \mathrm{d}^4x \left[-\frac{1}{2}\partial_\mu\phi(x)\partial^\mu\phi(x) - \frac{\lambda}{4!}\phi^4(x)\right] = S[\phi] \tag{3.5}$$

Now, let's compute the conserved charge,

$$J^\mu = \frac{\partial\mathcal{L}}{\partial\partial_\mu\phi(x)}\left.\frac{\mathrm{d}\phi'(x)}{\mathrm{d}a}\right|_{a=0} + \mathcal{L}\left.\frac{\mathrm{d}x'^\mu}{\mathrm{d}a}\right|_{a=0} \tag{3.6}$$

$$J^\mu = -\frac{\partial\mathcal{L}}{\partial\partial_\mu\phi(x)}(d_\phi\phi(x) + \partial_\alpha\phi(x)x^\alpha) + \mathcal{L}x^\mu \tag{3.7}$$

$$J^\mu = \partial^\mu\phi(d_\phi\phi + \partial_\alpha\phi x^\alpha) + \left(-\frac{1}{2}\partial^\alpha\phi\partial_\alpha\phi + \frac{\lambda}{4!}\phi^4\right)x^\mu \tag{3.8}$$

Check if it is really divergenceless,

$$\begin{aligned}\partial_\mu J^\mu = {} & \partial_\mu\partial^\mu\phi(d_\phi\phi + \partial_\alpha\phi x^\alpha) + \partial^\mu\phi(d_\phi\partial_\mu\phi + \partial_\mu\partial_\alpha\phi x^\alpha + \partial_\alpha\phi\partial_\mu x^\alpha) \\ & + \left(-\partial^\alpha\phi\partial_\mu\partial_\alpha\phi - \frac{\lambda}{3!}\phi^3\partial_\mu\phi\right)x^\mu + \left(-\frac{1}{2}\partial^\alpha\phi\partial_\alpha\phi - \frac{\lambda}{4!}\phi^4\right)\partial_\mu x^\mu\end{aligned} \tag{3.9}$$

$$\begin{aligned}\partial_\mu J^\mu = {} & \Box\phi(d_\phi\phi + \partial_\alpha\phi x^\alpha) + \partial^\mu\phi\big(d_\phi\partial_\mu\phi + \partial_\mu\partial_\alpha\phi x^\alpha + \partial_\alpha\phi g_\mu{}^\alpha\big) \\ & + \left(-\partial^\alpha\phi\partial_\mu\partial_\alpha\phi - \frac{\lambda}{3!}\phi^3\partial_\mu\phi\right)x^\mu + 4\left(-\frac{1}{2}\partial^\alpha\phi\partial_\alpha\phi - \frac{\lambda}{4!}\phi^4\right)\end{aligned} \tag{3.10}$$

$$\begin{aligned}\partial_\mu J^\mu = {} & \left(\Box\phi - \frac{\lambda}{3!}\phi^3\right)x^\alpha\partial_\alpha\phi + \left(d_\phi\Box\phi - \frac{\lambda}{3!}\phi^3\right)\phi + \partial^\mu\phi(d_\phi\partial_\mu\phi + \partial_\mu\partial_\alpha\phi x^\alpha + \partial_\mu\phi) \\ & - \partial^\alpha\phi\partial_\mu\partial_\alpha\phi x^\mu - 2\partial^\alpha\phi\partial_\alpha\phi\end{aligned} \tag{3.11}$$

$$\partial_\mu J^\mu = \left(\Box\phi - \frac{\lambda}{3!}\phi^3\right)x^\alpha\partial_\alpha\phi + \left(\Box\phi - \frac{\lambda}{3!}\phi^3\right)\phi + (d_\phi - 1)\phi\Box\phi + (d_\phi - 1)\partial^\mu\phi\partial_\mu\phi \tag{3.12}$$

Using the equation of motion,

$$\Box\phi = \frac{\lambda}{3!}\phi^3 \tag{3.13}$$

We get,

$$\partial_\mu J^\mu = (d_\phi - 1)\phi\Box\phi + (d_\phi - 1)\partial^\mu\phi\partial_\mu\phi \tag{3.14}$$

Now set $d_\phi = 1$,

$$\partial_\mu J^\mu = 0 \tag{3.15}$$

Problem 4

4.A)

The Lorentz transformation is,

$$x'^{\mu} = \Lambda^{\mu}{}_{\nu} x^{\nu} = x^{\mu} + \omega^{\mu}{}_{\nu} x^{\nu} + \mathcal{O}\big(\omega^2\big) \tag{4.1}$$

Which has to satisfy,

$$\begin{aligned}
y^{\mu} x_{\mu} &= y'^{\mu} x'_{\mu} && (4.2)\\
y^{\mu} x_{\mu} &= \big(y^{\mu} + \omega^{\mu}{}_{\nu} y^{\nu} + \mathcal{O}\big(\omega^2\big)\big)\big(x_{\mu} + \omega_{\mu\alpha} x^{\alpha} + \mathcal{O}\big(\omega^2\big)\big) && (4.3)\\
y^{\mu} x_{\mu} &= y^{\mu} x_{\mu} + \omega^{\mu}{}_{\nu} y^{\nu} x_{\mu} + y^{\mu} \omega_{\mu\alpha} x^{\alpha} + \mathcal{O}\big(\omega^2\big) && (4.4)\\
0 &= \omega_{\mu\nu} y^{\nu} x^{\mu} + \omega_{\mu\nu} x^{\nu} y^{\mu} && (4.5)\\
0 &= y^{\nu} x^{\mu} \big(\omega_{\mu\nu} + \omega_{\nu\mu}\big) \Rightarrow \omega_{\mu\nu} + \omega_{\nu\mu} = 0 \Rightarrow \omega_{\mu\nu} = -\omega_{\nu\mu} && (4.6)
\end{aligned}$$

4.B)

A rotation should preserve the time component,

$$\begin{aligned}
x^0 = x'^0 &= x^0 + \omega^0{}_{\nu} x^{\nu} && (4.7)\\
0 &= \omega^0{}_{\nu} x^{\nu} \Rightarrow \omega^0{}_{\nu} = 0 \Rightarrow \omega_{0\nu} = 0 && (4.8)
\end{aligned}$$

Together with the anti-symmetry gives,

$$\omega_{\mu\nu} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & \omega_{12} & -\omega_{31} \\ 0 & -\omega_{12} & 0 & \omega_{23} \\ 0 & \omega_{31} & -\omega_{23} & 0 \end{pmatrix} \tag{4.9}$$

4.C)

The Lorentz transformation for a scalar field is,

$$\begin{cases} x^{\mu} & \to x'^{\mu} = \Lambda^{\mu}{}_{\nu}(\omega) x^{\nu} \\ \phi(x) & \to \phi'(x') = \phi(x) \end{cases} \tag{4.10}$$

Which is parametrized by $\omega_{\mu\nu}$, thus,

$$\begin{aligned}
\left.\frac{\mathrm{d}x'^{\mu}}{\mathrm{d}\omega^{\alpha\beta}}\right|_{\omega_{\alpha\beta}=0} &= \left.\frac{\mathrm{d}}{\mathrm{d}\omega_{\alpha\beta}}\big(x^{\mu} + \omega^{\mu}{}_{\nu} x^{\nu} + \mathcal{O}\big(\omega^2\big)\big)\right|_{\omega_{\alpha\beta}=0} && (4.11)\\
\left.\frac{\mathrm{d}x'^{\mu}}{\mathrm{d}\omega^{\alpha\beta}}\right|_{\omega_{\alpha\beta}=0} &= \big(g_{\alpha}{}^{\mu} g_{\beta\nu} - g_{\beta}{}^{\mu} g_{\alpha\nu}\big) x^{\nu} = g_{\alpha}{}^{\mu} x_{\beta} - g_{\beta}{}^{\mu} x_{\alpha} && (4.12)
\end{aligned}$$

And,

$$\frac{\mathrm{d}\phi'(x)}{\mathrm{d}\omega^{\alpha\beta}}\bigg|_{\omega^{\alpha\beta}=0} = \frac{\mathrm{d}}{\mathrm{d}\omega^{\alpha\beta}}\big(\phi\big(\Lambda^{-1}x\big)\big)\bigg|_{\omega^{\alpha\beta}=0} = \partial_\nu\phi(x)\frac{\mathrm{d}}{\mathrm{d}\omega^{\alpha\beta}}\big(\Lambda^{-1}x\big)^\nu\bigg|_{\omega^{\alpha\beta}=0} \tag{4.13}$$

$$\frac{\mathrm{d}\phi'(x)}{\mathrm{d}\omega^{\alpha\beta}}\bigg|_{\omega^{\alpha\beta}=0} = \partial_\nu\phi(x)\frac{\mathrm{d}\Lambda_\rho{}^\nu}{\mathrm{d}\omega^{\alpha\beta}}\bigg|_{\omega^{\alpha\beta}=0} x^\rho = \partial_\nu\phi(x)\big(g_{\alpha\rho}g_\beta{}^\nu - g_{\beta\rho}g_\alpha{}^\nu\big)x^\rho \tag{4.14}$$

$$\frac{\mathrm{d}\phi'(x)}{\mathrm{d}\omega^{\alpha\beta}}\bigg|_{\omega^{\alpha\beta}=0} = x_\alpha\partial_\beta\phi - x_\beta\partial_\alpha\phi \tag{4.15}$$

Using these results in eq. (D.21),

$$J^\mu{}_{\alpha\beta} = \frac{\partial\mathcal{L}}{\partial\partial_\mu\phi(x)}\frac{\mathrm{d}\phi'(x)}{\mathrm{d}\omega^{\alpha\beta}}\bigg|_{\omega^{\alpha\beta}=0} + \mathcal{L}\frac{\mathrm{d}x'^\mu}{\mathrm{d}\omega^{\alpha\beta}}\bigg|_{\omega^{\alpha\beta}=0} \tag{4.16}$$

$$J^\mu{}_{\alpha\beta} = \frac{\partial\mathcal{L}}{\partial\partial_\mu\phi(x)}(x_\alpha\partial_\beta\phi - x_\beta\partial_\alpha\phi) + \mathcal{L}\big(g_\alpha{}^\mu x_\beta - g_\beta{}^\mu x_\alpha\big) \tag{4.17}$$

$$J^\mu{}_{\alpha\beta} = x_\beta J^\mu{}_\alpha - x_\alpha J^\mu{}_\beta \tag{4.18}$$

Check that is conserved,

$$\partial_\mu J^\mu{}_{\alpha\beta} = \partial_\mu\big(x_\beta J^\mu{}_\alpha - x_\alpha J^\mu{}_\beta\big) \tag{4.19}$$

$$\partial_\mu J^\mu{}_{\alpha\beta} = \partial_\mu x_\beta J^\mu{}_\alpha + x_\beta\partial_\mu J^\mu{}_\alpha - \partial_\mu x_\alpha J^\mu{}_\beta - x_\alpha\partial_\mu J^\mu{}_\beta \tag{4.20}$$

$$\partial_\mu J^\mu{}_{\alpha\beta} = g_{\mu\beta}J^\mu{}_\alpha + x_\beta\partial_\mu J^\mu{}_\alpha - g_{\mu\alpha}J^\mu{}_\beta - x_\alpha\partial_\mu J^\mu{}_\beta \tag{4.21}$$

$$\partial_\mu J^\mu{}_{\alpha\beta} = J_{\beta\alpha} - J_{\alpha\beta} + x_\beta\partial_\mu J^\mu{}_\alpha - x_\alpha\partial_\mu J^\mu{}_\beta \tag{4.22}$$

Without a proper lagrangian this is as far as we can go, but, an interesting result that can be seen from here is, whenever the energy momentum tensor is conserved and symmetric, the angular momentum tensor is conserved.

Problem 5

The Lorentz transformation for a vector field is,

$$\begin{cases} x^\mu & \to x'^\mu = \Lambda^\mu{}_\nu(\omega)x^\nu \\ A^\rho(x) & \to A'^\rho(x') = \Lambda^\rho{}_\sigma A^\sigma(x) \end{cases} \tag{5.1}$$

The first step we had already made, eq. (4.12),

$$\left.\frac{\mathrm{d}x'^\mu}{\mathrm{d}\omega^{\alpha\beta}}\right|_{\omega_{\alpha\beta}=0} = g_\alpha{}^\mu x_\beta - g_\beta{}^\mu x_\alpha \tag{5.2}$$

The second step,

$$\left.\frac{\mathrm{d}A'^\rho(x)}{\mathrm{d}\omega^{\alpha\beta}}\right|_{\omega^{\alpha\beta}=0} = \left.\frac{\mathrm{d}}{\mathrm{d}\omega^{\alpha\beta}}\big(\Lambda^\rho{}_\sigma A^\sigma\big(\Lambda^{-1}x\big)\big)\right|_{\omega^{\alpha\beta}=0} \tag{5.3}$$

$$\left.\frac{\mathrm{d}A'^\rho(x)}{\mathrm{d}\omega^{\alpha\beta}}\right|_{\omega^{\alpha\beta}=0} = \left.\frac{\mathrm{d}}{\mathrm{d}\omega^{\alpha\beta}}(\Lambda^\rho{}_\sigma)\right|_{\omega^{\alpha\beta}=0} A^\sigma(x) + g^\rho{}_\sigma \left.\frac{\mathrm{d}}{\mathrm{d}\omega^{\alpha\beta}}\big(A^\sigma\big(\Lambda^{-1}x\big)\big)\right|_{\omega^{\alpha\beta}=0} \tag{5.4}$$

$$\left.\frac{\mathrm{d}A'^\rho(x)}{\mathrm{d}\omega^{\alpha\beta}}\right|_{\omega^{\alpha\beta}=0} = \big(g_\alpha{}^\rho g_{\beta\sigma} - g_\beta{}^\rho g_{\alpha\sigma}\big)A^\sigma(x) + \partial_\nu A^\rho(x)\left.\frac{\mathrm{d}}{\mathrm{d}\omega^{\alpha\beta}}\big(\Lambda^{-1}x\big)^\nu\right|_{\omega^{\alpha\beta}=0} \tag{5.5}$$

$$\left.\frac{\mathrm{d}A'^\rho(x)}{\mathrm{d}\omega^{\alpha\beta}}\right|_{\omega^{\alpha\beta}=0} = g_\alpha{}^\rho A_\beta - g_\beta{}^\rho A_\alpha + \partial_\nu A^\rho(x)\left.\frac{\mathrm{d}\Lambda_\gamma{}^\nu}{\mathrm{d}\omega^{\alpha\beta}}\right|_{\omega^{\alpha\beta}=0} x^\gamma \tag{5.6}$$

$$\left.\frac{\mathrm{d}A'^\rho(x)}{\mathrm{d}\omega^{\alpha\beta}}\right|_{\omega^{\alpha\beta}=0} = g_\alpha{}^\rho A_\beta - g_\beta{}^\rho A_\alpha + \partial_\nu A^\rho(x)\big(g_{\alpha\gamma}g_\beta{}^\nu - g_{\beta\gamma}g_\alpha{}^\nu\big)x^\gamma \tag{5.7}$$

$$\left.\frac{\mathrm{d}A'^\rho(x)}{\mathrm{d}\omega^{\alpha\beta}}\right|_{\omega^{\alpha\beta}=0} = g_\alpha{}^\rho A_\beta - g_\beta{}^\rho A_\alpha + x_\alpha\partial_\beta A^\rho(x) - x_\beta\partial_\alpha A^\rho(x) \tag{5.8}$$

$$\tag{5.9}$$

Using these results in eq. (D.21),

$$J^\mu{}_{\alpha\beta} = \frac{\partial\mathcal{L}}{\partial\partial_\mu A^\rho(x)}\left.\frac{\mathrm{d}A'^\rho(x)}{\mathrm{d}\omega^{\alpha\beta}}\right|_{\omega^{\alpha\beta}=0} + \mathcal{L}\left.\frac{\mathrm{d}x'^\mu}{\mathrm{d}\omega^{\alpha\beta}}\right|_{\omega^{\alpha\beta}=0} \tag{5.10}$$

$$J^\mu{}_{\alpha\beta} = \frac{\partial\mathcal{L}}{\partial\partial_\mu A^\rho(x)}\big(g_\alpha{}^\rho A_\beta - g_\beta{}^\rho A_\alpha + x_\alpha\partial_\beta A^\rho - x_\beta\partial_\alpha A^\rho\big) + \mathcal{L}\big(g_\alpha{}^\mu x_\beta - g_\beta{}^\mu x_\alpha\big) \tag{5.11}$$

$$J^\mu{}_{\alpha\beta} = x_\beta J^\mu{}_\alpha - x_\alpha J^\mu{}_\beta + \frac{\partial\mathcal{L}}{\partial\partial_\mu A^\alpha(x)}A_\beta - \frac{\partial\mathcal{L}}{\partial\partial_\mu A^\beta(x)}A_\alpha \tag{5.12}$$

Which also isn't gauge invariant. To solve this problem consider the following,

$$J^\mu{}_{\alpha\beta} = x_\beta J^\mu{}_\alpha - x_\alpha J^\mu{}_\beta + S^\mu{}_{\alpha\beta} \tag{5.13}$$

The conservation equation gives,

$$\partial_\mu J^\mu{}_{\alpha\beta} = J_{\beta\alpha} - J_{\alpha\beta} + \partial_\mu S^\mu{}_{\alpha\beta} = 0 \tag{5.14}$$

$$\partial_\mu S^\mu{}_{\alpha\beta} = J_{\alpha\beta} - J_{\beta\alpha} \tag{5.15}$$

Thus, we can get from $J_{\alpha\beta}$ a symmetric tensor by summing a divergenceless tensor,

$$\Theta_{\alpha\beta} = J_{\alpha\beta} + \frac{1}{2}\big(J_{\alpha\beta} - J_{\beta\alpha}\big) - \frac{1}{2}\partial\big(J_{\alpha\beta} - J_{\beta\alpha}\big) = J_{\alpha\beta} + \frac{1}{2}\partial_\mu S^\mu{}_{\alpha\beta} \tag{5.16}$$

A Conventions

A.A) Metric

There are two choices of metric,

$$g_- = \mathrm{diag}\big(- \quad + \quad + \quad +\big), \quad g_+ = \mathrm{diag}\big(+ \quad - \quad - \quad -\big) \tag{A.1}$$

I will always use g_-. This changes a lot of signs in a lot of places, as example,

$$\mathcal{L} = g_+^{\mu\nu}\frac{1}{2}\partial_\mu\phi\partial_\nu\phi - V[\phi] = -g_-^{\mu\nu}\frac{1}{2}\partial_\mu\phi\partial_\nu\phi - V[\phi] \tag{A.2}$$

Physics should be invariant under conventions, so they don't matter, as long as you know which conventions you are using and are consistent.

A.B) Levi-Civita

The Levi-Civita is the (unique) totally antisymmetric four-pseudo-tensor, $\epsilon_{\alpha\beta\mu\nu}$, there are two possible choices of normalization,

$$\epsilon^+_{0123} = 1, \quad \epsilon^-_{0123} = -1 \tag{A.3}$$

I will always use ϵ^+.

B Lorentz Group Digression

The Lorentz group is defined as the set of all 4x4 matrices which preserve the scalar products with respect to the relativistic metric, that is, the set of all $\Lambda : x^\mu \to x'^\mu = \Lambda^\mu{}_\nu x^\nu$, such,

$$x^\mu g_{\mu\nu} y^\nu = x'^\alpha g_{\alpha\beta} y'^\beta \tag{B.1}$$
$$x^\mu g_{\mu\nu} y^\nu = \Lambda^\alpha{}_\mu x^\mu g_{\alpha\beta} y^\nu \Lambda^\beta{}_\nu \tag{B.2}$$
$$x^\mu g_{\mu\nu} y^\nu = x^\mu y^\nu \Lambda^\alpha{}_\mu g_{\alpha\beta} \Lambda^\beta{}_\nu \tag{B.3}$$

as this should be valid for all x, y,

$$g_{\mu\nu} = \Lambda^{\mathrm{T}}{}_\mu{}^\alpha g_{\alpha\beta} \Lambda^\beta{}_\nu \tag{B.4}$$
$$g = \Lambda^{\mathrm{T}} g \Lambda \tag{B.5}$$

in other words, the Lorentz group is the group of all orthogonal 4x4 matrices with respect to the non-positive definite metric g. For orthogonal nxn matrices with respect to the standard $\mathbb{R}^n$ metric, this group is denoted $O(n)$, as we're in a 1 negative and 3 positive eigenvalues metric, we denote by $O(1,3) \cong O(3,1)$. Apart from the topological structure carried by $O(1,3)$, we are mainly interested in the tangent space structure at the origin, at the identity element, also know as the Lie algebra, $\mathfrak{o}(1,3)$. This is the case because the generators of the Lie algebra are in a one to one correspondence with hermitian operators in the Hilbert space of the quantum theory which represents conserved charges, for the Lorentz group these are the angular momentum and boost operators. It's not hard to obtain the connected piece of $O(1,3)$, we have two conditions which splits our group into 4 connected pieces,

$$\mathrm{Det}[g] = \mathrm{Det}[\Lambda^{\mathrm{T}}]\mathrm{Det}[g]\mathrm{Det}[\Lambda] \tag{B.6}$$
$$-1 = -\mathrm{Det}[\Lambda^{\mathrm{T}}]\mathrm{Det}[\Lambda] \tag{B.7}$$
$$1 = (\mathrm{Det}[\Lambda])^2 \Rightarrow \mathrm{Det}[\Lambda] = \pm 1 \tag{B.8}$$

and also,

$$g_{00} = \Lambda^{\mathrm{T}}{}_0{}^\alpha g_{\alpha\beta} \Lambda^\beta{}_0 \tag{B.9}$$
$$-1 = \Lambda^{\mathrm{T}}{}_0{}^0 g_{00} \Lambda^0{}_0 + \Lambda^{\mathrm{T}}{}_0{}^i g_{ij} \Lambda^j{}_0 \tag{B.10}$$
$$\Lambda^0{}_0 \Lambda^0{}_0 = 1 + \sum_i \left\| \Lambda^i{}_0 \right\|^2 \geq 1 \tag{B.11}$$
$$\left(\Lambda^0{}_0\right)^2 \geq 1 \Rightarrow \Lambda^0{}_0 \begin{cases} \geq 1 \\ \leq -1 \end{cases} \tag{B.12}$$

The connected piece is the one satisfying $\mathrm{Det}[\Lambda] = +1$ and $\Lambda^0{}_0 \geq 1$, which is called the proper orthochronous Lorentz group $SO^+(1,3)$, proper (S) for $\mathrm{Det}[\Lambda] = +1$ and orthochronous $(+)$ for $\Lambda^0{}_0 \geq 1$. It's not hard to see that from the connected piece we can get all the other three pieces by compositions with the following matrices,

$$\mathcal{P}^\mu{}_\nu = \mathrm{diag}\begin{pmatrix}1 & -1 & -1 & -1\end{pmatrix}, \quad \mathcal{T}^\mu{}_\nu = \mathrm{diag}\begin{pmatrix}-1 & 1 & 1 & 1\end{pmatrix} \tag{B.13}$$
$$[\mathcal{P}\mathcal{T}]^\mu{}_\nu = \mathrm{diag}\begin{pmatrix}-1 & -1 & -1 & -1\end{pmatrix} \tag{B.14}$$

in fact they form a group by themselves, $\{\mathbb{1}, \mathcal{P}, \mathcal{T}, \mathcal{P}\mathcal{T}\}$, called the Klein four group. With them we can reconstruct the full Lorentz group,

$$O(1,3) = \{\mathbb{1}, \mathcal{P}, \mathcal{T}, \mathcal{P}\mathcal{T}\} \rtimes SO^+(1,3) \tag{B.15}$$

The quantum theory allows for projective representations of groups, this is due to states of the Hilbert space which differ by a phase represent the same physical configuration, hence, representations of groups can obey the group multiplication law up to a phase, a.k.a. projective representations. Constructing all the projective representations is similar to constructing all the proper representations of a cover of the group, for quantum mechanics the double cover is enough. It's known that the double cover of $SO^+(1,3)$ is $SL(2,\mathbb{C})$, the group of 2x2 matrices of unit determinant over the complex numbers, that is,

$$SO^+(1,3) \cong SL(2,\mathbb{C})/\mathbb{Z}_2 \tag{B.16}$$

the real quantum symmetry group is then $SL(2,\mathbb{C})$, due to it being the double cover, it contains proper spinorial representations, which are not present at $SO^+(1,3)$, at least as proper representations.

Let's construct this isomorphism, first, a disclaimer, we'll use a little bit of a posteriori knowledge, which could be obtained through the item 1.*b* without knowing of this construction. What we need is a way to represent Lorentz vectors over the 2x2 complex values matrices, in which a proper orthochronous Lorentz transformation is given through a unit determinant transformation. The correct choice, which we'll leave to explain why latter,

$$X = \begin{pmatrix} -x^0 + x^3 & x^1 - \mathrm{i}x^2 \\ x^1 + \mathrm{i}x^2 & -x^0 - x^3 \end{pmatrix} = -x^0\mathbb{1} + x^i\sigma_i = x^\mu\sigma_\mu, \quad \mathrm{Det}[X] = -x_\mu x^\mu \tag{B.17}$$

with σ_i being the usual Pauli matrices, we also defined $\sigma_0 = -\mathbb{1}$, we leave for now open as if this is an honest four vector, but in fact it transforms covariantly over $SL(2,\mathbb{C})$, and even more, it's invariant. As the scalar product is given by the determinant, Lorentz transformations are linear matrix multiplications which preserve the determinant,

$$X' = \begin{pmatrix} -x'^0 + x'^3 & x'^1 - \mathrm{i}x'^2 \\ x'^1 + \mathrm{i}x'^2 & -x'^0 - x'^3 \end{pmatrix} = \tilde{U}X\tilde{V}, \quad \mathrm{Det}[X'] = \mathrm{Det}[X] \tag{B.18}$$

also, the transformation needs to preserve the reality of each of the components, this can be assured by noting that $X^\dagger = X$, and then demanding that $X'^\dagger = X'$, which fixes $\tilde{V} = \tilde{U}^\dagger$. The preservation of the determinant gives the following constraint,

$$\mathrm{Det}[X] = \mathrm{Det}[X'] \tag{B.19}$$

$$\mathrm{Det}[X] = \mathrm{Det}[\tilde{U}]\mathrm{Det}[X]\mathrm{Det}[\tilde{U}^\dagger] \tag{B.20}$$

$$1 = \left\|\mathrm{Det}[\tilde{U}]\right\|^2 \Rightarrow \mathrm{Det}[\tilde{U}] = \exp(\mathrm{i}\alpha) \tag{B.21}$$

notice that $\tilde{U}$ and $U = \exp\left(-\mathrm{i}\frac{\alpha}{2}\right)\tilde{U}$ generates the same transformation on X, hence, we can consider only the determinant one matrices. What we concluded up to now is that at least the 2x2 unit determinant matrices over the complexes is surjective in the (proper orthochronous?) Lorentz group, let's verify it's the proper orthochronous, to see the condition ${\Lambda^0}_0 \geq 1$, take $X = -\mathbb{1} \leftrightarrow x = \mathrm{diag}\begin{pmatrix}1 & 0 & 0 & 0\end{pmatrix}$, then,

$$X' = \begin{pmatrix} -{\Lambda^0}_0 + {\Lambda^3}_0 & {\Lambda^1}_0 - \mathrm{i}{\Lambda^2}_0 \\ {\Lambda^1}_0 + \mathrm{i}{\Lambda^2}_0 & -{\Lambda^0}_0 - {\Lambda^3}_0 \end{pmatrix} \Rightarrow \mathrm{Tr}[X'] = -2{\Lambda^0}_0 \tag{B.22}$$

thus,

$${\Lambda^0}_0 = -\frac{1}{2}\mathrm{Tr}[X'] = -\frac{1}{2}\mathrm{Tr}\left[UXU^\dagger\right] = \frac{1}{2}\mathrm{Tr}\left[UU^\dagger\right] \tag{B.23}$$

as $\mathrm{Det}[U] = 1$, we have $\mathrm{Det}[UU^\dagger] = 1$, if λ_1, λ_2 are the eigenvalues of U, λ_1^*, λ_2^* are the eigenvalues of $U^\dagger$, hence, the eigenvalues of $UU^\dagger$ are $\|\lambda_1\|^2, \|\lambda_2\|^2 > 0$, the constraint $\mathrm{Det}[UU^\dagger] = 1 = \|\lambda_1\|^2\|\lambda_2\|^2$ imposes that $\mathrm{Tr}\big[UU^\dagger\big] = \|\lambda_1\|^2 + \|\lambda_1\|^2 \geq 2$, therefore,

$$\Lambda^0{}_0 = \frac{1}{2}\,\mathrm{Tr}\big[UU^\dagger\big] \geq 1 \tag{B.24}$$

this clears that $SL(2,\mathbb{C})$ is at least surjective in the orthochronous Lorentz group. Now, the proper part, it suffices to show that there isn't in this group a parity transformation. Suppose exists U_p such that,

$$X' = U_p X U_p'^\dagger = U_p\big(-x^0\mathbb{1} + x^i\sigma_i\big)U_p^\dagger = -x^0\mathbb{1} - x^i\sigma_i \tag{B.25}$$

which imposes,

$$U_p U_p^\dagger = \mathbb{1} \tag{B.26}$$

$$U_p \sigma_i U_p^\dagger = -\sigma_i \tag{B.27}$$

Now, $\sigma_1\sigma_2\sigma_3 = \mathrm{i}\mathbb{1}$ (check it!), thus, we can compute the following,

$$U_p\sigma_1 U_p^\dagger U_p \sigma_2 U_p^\dagger U_p \sigma_3 U_p^\dagger = (-)^3\sigma_1\sigma_2\sigma_3 \tag{B.28}$$

$$U_p\sigma_1\sigma_2\sigma_3 U_p^\dagger = -\mathrm{i}\mathbb{1} \tag{B.29}$$

$$U_p \mathrm{i}\mathbb{1} U_p^\dagger = -\mathrm{i}\mathbb{1} \tag{B.30}$$

$$U_p U_p^\dagger = -\mathbb{1} \tag{B.31}$$

contradiction! Thus the parity transformation with $\mathrm{Det}[\mathcal{P}] = -1$ isn't present in $SL(2,\mathbb{C})$, and as this is a connected continuous group, none of the transformations with $\mathrm{Det}[\Lambda] = -1$ are present. This concludes that $SL(2,\mathbb{C})$ is surjective over the proper orthochronous Lorentz group. But is still up to discussion whether this is an injective map, the answer is, it isn't, more than one transformation in $SL(2,\mathbb{C})$ correspond to the same $SO^+(1,3)$ transformation. This can be seen from the transformation law,

$$X' = UXU^\dagger = (-U)X(-U)^\dagger \tag{B.32}$$

as U is two dimensional, $\mathrm{Det}[-U] = 1$, and we conclude that both U and $-U$ generate the same proper orthochronous Lorentz transformation, formally this means that $SO^+(1,3)$ can be seen as an equivalence class over $SL(2,\mathbb{C})$,

$$SO^+(1,3) \cong \{U \in SL(2,\mathbb{C}) | U \sim -U\} \tag{B.33}$$

the action $U \to -U$ can be seen as a $\mathbb{Z}_2$ action over $SL(2,\mathbb{C})$, hence, this equivalence class is a quotient group,

$$SO^+(1,3) \cong SL(2,\mathbb{C})/\mathbb{Z}_2 \tag{B.34}$$

this also means that $\mathfrak{o}(1,3) \cong \mathfrak{so}(1,3) \cong \mathfrak{so}^+(1,3) \cong \mathfrak{sl}(2,\mathbb{C})$. The presence of $\mathbb{Z}_2$ instead of $\mathbb{Z}_n$ signals that $SL(2,\mathbb{C})$ is the double-cover, actually in this case it is the universal-cover. For general dimensions the double-cover of $SO^+(p,q)$ is defined as $\mathrm{Spin}^+(p,q)$,

$$SO^+(p,q) \cong \mathrm{Spin}^+(p,q)/\mathbb{Z}_2 \tag{B.35}$$

in a comically similar fashion, we have the following,

$$SO(p,q) \cong \mathrm{Spin}(p,q)/\mathbb{Z}_2, \quad O(p,q) \cong \mathrm{Pin}(p,q)/\mathbb{Z}_2 \tag{B.36}$$

these groups $\mathrm{Spin}^+(p,q), \mathrm{Spin}(p,q), \mathrm{Pin}(p,q)$ are more easily definable by means of Clifford algebras, which we'll see soon enough, a representation of the Clifford algebra $\mathrm{Cl}(p,q)$ is given by the Dirac gamma matrices. The interesting point is that $\mathrm{Spin}(1,3) \cong \mathrm{Spin}(3,1)$ contains the $\mathcal{PT}$ transformation, but not parity and time inversion separately, but the Pin group, which contains all $\mathcal{P}, \mathcal{T}, \mathcal{PT}$ has the incredible property of

$$\mathrm{Pin}(1,3) \ncong \mathrm{Pin}(3,1) \tag{B.37}$$

which is a consequence of a non isomorphism between the Clifford algebras,

$$\mathrm{Cl}(1,3) \ncong \mathrm{Cl}(3,1) \tag{B.38}$$

This is simply outstanding, which in principle means that in a quantum theory with fermions and in which all $\mathcal{P}, \mathcal{T}, \mathcal{PT}$ are true symmetries there could be noticeable experimental results that could decide in favor of $\mathrm{Pin}(3,1)$ over $\mathrm{Pin}(1,3)$.

C Determinants

D Noether's Theorem

Here we're going to derive the Noether theorem for a handful of cases, first of all, the validity of this theorem is to global continuous symmetries, which in our field theory context means: we start of with an action which depends upon a field, $S[\phi]$, it's supposed to encode whatever index the fields have, then, a symmetry transformation is given in terms of:

$$\begin{cases} x^\mu & \to x'^\mu = x'^\mu(x;a) \\ \phi(x) & \to \phi'(x') = \phi'(\phi(x), x; a) \end{cases} \Rightarrow S[\phi] = S'_a[\phi'] \tag{D.1}$$

Let's look further at the symmetry condition,

$$S'_a[\phi'] - S[\phi] = \int\limits_{\Omega'} \mathrm{d}^4x'\, \mathcal{L}'[\phi'(x'); x'] - \int\limits_{\Omega} \mathrm{d}^4x\, \mathcal{L}[\phi(x); x] \tag{D.2}$$

$$\begin{aligned} S'_a[\phi'] - S[\phi] = &\int\limits_{\Omega} \mathrm{d}^4x\, \mathcal{L}'[\phi'(x); x] - \int\limits_{\Omega} \mathrm{d}^4x\, \mathcal{L}[\phi(x); x] \\ &+ \int\limits_{\Omega'} \mathrm{d}^4x'\, \mathcal{L}'[\phi'(x'); x'] - \int\limits_{\Omega} \mathrm{d}^4x\, \mathcal{L}'[\phi'(x); x] \end{aligned} \tag{D.3}$$

$$\begin{aligned} S'_a[\phi'] - S[\phi] = &\int\limits_{\Omega} \mathrm{d}^4x\, \{\mathcal{L}'[\phi'(x); x] - \mathcal{L}[\phi(x); x]\} \\ &+ \int\limits_{\Omega} \mathrm{d}^4x \left\{ \mathrm{Det}\left[\frac{\partial x'}{\partial x}\right] \mathcal{L}'[\phi'(x'); x'] - \mathcal{L}'[\phi'(x); x] \right\} \end{aligned} \tag{D.4}$$

$$\begin{aligned} S'_a[\phi'] - S[\phi] = &\int\limits_{\Omega} \mathrm{d}^4x\, \{\mathcal{L}'[\phi'(x); x] - \mathcal{L}[\phi(x); x]\} \\ &+ \int\limits_{\Omega} \mathrm{d}^4x \left\{ \mathrm{Det}\left[\frac{\partial x'}{\partial x}\right] - 1 \right\} \mathcal{L}'[\phi'(x'); x'] \\ &+ \int\limits_{\Omega} \mathrm{d}^4x\, \{\mathcal{L}'[\phi'(x'); x'] - \mathcal{L}'[\phi'(x); x]\} \end{aligned} \tag{D.5}$$

Suppose $\mathcal{L}' = \mathcal{L}$,

$$\begin{aligned} \left.\frac{[S'_a[\phi'] - S[\phi]]}{a}\right|_{a\to 0} = &\int\limits_{\Omega} \mathrm{d}^4x \int\limits_{\Omega} \mathrm{d}^4y \left.\left\{ \frac{\delta\mathcal{L}}{\delta\phi(y)} \frac{\phi'(y) - \phi(y)}{a} \right\}\right|_{a\to 0} \\ &+ \int\limits_{\Omega} \mathrm{d}^4x\, \frac{1}{a} \left.\left\{ \mathrm{Det}\left[\frac{\partial x'}{\partial x}\right] - 1 \right\}\right|_{a\to 0} \mathcal{L} \\ &+ \int\limits_{\Omega} \mathrm{d}^4x\, \frac{\mathrm{d}\mathcal{L}}{\mathrm{d}x^\alpha} \left.\frac{x'^\alpha - x^\alpha}{a}\right|_{a\to 0} \end{aligned}$$

$$\begin{aligned} \left.\frac{\mathrm{d}}{\mathrm{d}a} S'_a[\phi']\right|_{a=0} = &\int\limits_{\Omega} \mathrm{d}^4x \int\limits_{\Omega} \mathrm{d}^4y\, \left.\frac{\delta\mathcal{L}}{\delta\phi(y)} \frac{\mathrm{d}\phi'(y)}{\mathrm{d}a}\right|_{a=0} \\ &+ \int\limits_{\Omega} \mathrm{d}^4x\, \mathcal{L} \frac{\mathrm{d}}{\mathrm{d}a} \left.\mathrm{Det}\left[\frac{\partial x'}{\partial x}\right]\right|_{a=0} \end{aligned}$$

$$+\int_\Omega \mathrm{d}^4x\,\frac{\mathrm{d}\mathcal{L}}{\mathrm{d}x^\alpha}\frac{\mathrm{d}x'^\alpha}{\mathrm{d}a}\bigg|_{a=0} \tag{D.6}$$

Suppose the lagrangian has at most second derivatives,

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}a}S'_a[\phi']\bigg|_{a=0} &= \int_\Omega \mathrm{d}^4x \int_\Omega \mathrm{d}^4y \left[\frac{\partial\mathcal{L}}{\partial\phi(x)}\delta^{(4)}(x-y) + \frac{\partial\mathcal{L}}{\partial\partial_{x\mu}\phi(x)}\partial_{x\mu}\delta^{(4)}(x-y)\right]\frac{\mathrm{d}\phi'(y)}{\mathrm{d}a}\bigg|_{a=0} \\ &\quad + \int_\Omega \mathrm{d}^4x\,\mathcal{L}\frac{\mathrm{d}}{\mathrm{d}a}\mathrm{Det}\left[\frac{\partial x'}{\partial x}\right]\bigg|_{a=0} \\ &\quad + \int_\Omega \mathrm{d}^4x\,\frac{\mathrm{d}\mathcal{L}}{\mathrm{d}x^\alpha}\frac{\mathrm{d}x'^\alpha}{\mathrm{d}a}\bigg|_{a=0}\end{aligned} \tag{D.7}$$

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}a}S'_a[\phi']\bigg|_{a=0} &= \int_\Omega \mathrm{d}^4x \left[\frac{\partial\mathcal{L}}{\partial\phi(x)}\frac{\mathrm{d}\phi'(x)}{\mathrm{d}a}\bigg|_{a=0} + \frac{\partial\mathcal{L}}{\partial\partial_\mu\phi(x)}\partial_\mu\frac{\mathrm{d}\phi'(x)}{\mathrm{d}a}\bigg|_{a=0}\right] \\ &\quad + \int_\Omega \mathrm{d}^4x\,\mathcal{L}\frac{\mathrm{d}}{\mathrm{d}a}\mathrm{Det}\left[\frac{\partial x'}{\partial x}\right]\bigg|_{a=0} \\ &\quad + \int_\Omega \mathrm{d}^4x\,\frac{\mathrm{d}\mathcal{L}}{\mathrm{d}x^\alpha}\frac{\mathrm{d}x'^\alpha}{\mathrm{d}a}\bigg|_{a=0}\end{aligned} \tag{D.8}$$

Now impose the equations of motion,

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}a}S'_a[\phi']\bigg|_{a=0} &= \int_\Omega \mathrm{d}^4x \left[\partial_\mu\frac{\partial\mathcal{L}}{\partial\partial_\mu\phi(x)}\frac{\mathrm{d}\phi'(x)}{\mathrm{d}a}\bigg|_{a=0} + \frac{\partial\mathcal{L}}{\partial\partial_\mu\phi(x)}\partial_\mu\frac{\mathrm{d}\phi'(x)}{\mathrm{d}a}\bigg|_{a=0}\right] \\ &\quad + \int_\Omega \mathrm{d}^4x\,\mathcal{L}\frac{\mathrm{d}}{\mathrm{d}a}\mathrm{Det}\left[\frac{\partial x'}{\partial x}\right]\bigg|_{a=0} \\ &\quad + \int_\Omega \mathrm{d}^4x\,\frac{\mathrm{d}\mathcal{L}}{\mathrm{d}x^\alpha}\frac{\mathrm{d}x'^\alpha}{\mathrm{d}a}\bigg|_{a=0}\end{aligned} \tag{D.9}$$

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}a}S'_a[\phi']\bigg|_{a=0} &= \int_\Omega \mathrm{d}^4x\,\partial_\mu\left[\frac{\partial\mathcal{L}}{\partial\partial_\mu\phi(x)}\frac{\mathrm{d}\phi'(x)}{\mathrm{d}a}\bigg|_{a=0}\right] \\ &\quad + \int_\Omega \mathrm{d}^4x\,\mathcal{L}\frac{\mathrm{d}}{\mathrm{d}a}\mathrm{Det}\left[\frac{\partial x'}{\partial x}\right]\bigg|_{a=0} \\ &\quad + \int_\Omega \mathrm{d}^4x\,\frac{\mathrm{d}\mathcal{L}}{\mathrm{d}x^\alpha}\frac{\mathrm{d}x'^\alpha}{\mathrm{d}a}\bigg|_{a=0}\end{aligned} \tag{D.10}$$

The last missing piece is to compute the determinant,

$$\mathrm{Det}\left[\frac{\partial x'}{\partial x}\right] = -\frac{1}{4!}\epsilon_{\alpha\beta\gamma\delta}\epsilon^{\mu\nu\rho\sigma}\frac{\partial x'^\alpha}{\partial x^\mu}\frac{\partial x'^\beta}{\partial x^\nu}\frac{\partial x'^\gamma}{\partial x^\rho}\frac{\partial x'^\delta}{\partial x^\sigma} \tag{D.11}$$

$$\frac{\mathrm{d}}{\mathrm{d}a}\left[\mathrm{Det}\left[\frac{\partial x'}{\partial x}\right]\right]\bigg|_{a=0} = -\frac{1}{4!}\epsilon_{\alpha\beta\gamma\delta}\epsilon^{\mu\nu\rho\sigma}\frac{\mathrm{d}}{\mathrm{d}a}\left[\frac{\partial x'^\alpha}{\partial x^\mu}\frac{\partial x'^\beta}{\partial x^\nu}\frac{\partial x'^\gamma}{\partial x^\rho}\frac{\partial x'^\delta}{\partial x^\sigma}\right]\bigg|_{a=0} \tag{D.12}$$

$$\frac{\mathrm{d}}{\mathrm{d}a}\left[\mathrm{Det}\left[\frac{\partial x'}{\partial x}\right]\right]\bigg|_{a=0} = -\frac{1}{3!}\epsilon_{\alpha\beta\gamma\delta}\epsilon^{\mu\nu\rho\sigma}\frac{\mathrm{d}}{\mathrm{d}a}\left[\frac{\partial x'^{\alpha}}{\partial x^{\mu}}\right]\bigg|_{a=0}\frac{\partial x'^{\beta}}{\partial x^{\nu}}\bigg|_{a=0}\frac{\partial x'^{\gamma}}{\partial x^{\rho}}\bigg|_{a=0}\frac{\partial x'^{\delta}}{\partial x^{\sigma}}\bigg|_{a=0} \tag{D.13}$$

$$\frac{\mathrm{d}}{\mathrm{d}a}\left[\mathrm{Det}\left[\frac{\partial x'}{\partial x}\right]\right]\bigg|_{a=0} = -\frac{1}{3!}\epsilon_{\alpha\beta\gamma\delta}\epsilon^{\mu\nu\rho\sigma}\frac{\mathrm{d}}{\mathrm{d}a}\left[\frac{\partial x'^{\alpha}}{\partial x^{\mu}}\right]\bigg|_{a=0} g^{\beta}{}_{\nu}g^{\gamma}{}_{\rho}g^{\delta}{}_{\sigma} \tag{D.14}$$

$$\frac{\mathrm{d}}{\mathrm{d}a}\left[\mathrm{Det}\left[\frac{\partial x'}{\partial x}\right]\right]\bigg|_{a=0} = -\frac{1}{3!}\epsilon_{\alpha\nu\rho\sigma}\epsilon^{\mu\nu\rho\sigma}\frac{\mathrm{d}}{\mathrm{d}a}\left[\frac{\partial x'^{\alpha}}{\partial x^{\mu}}\right]\bigg|_{a=0} \tag{D.15}$$

$$\frac{\mathrm{d}}{\mathrm{d}a}\left[\mathrm{Det}\left[\frac{\partial x'}{\partial x}\right]\right]\bigg|_{a=0} = \frac{1}{3!}3!g_{\alpha}{}^{\mu}\frac{\mathrm{d}}{\mathrm{d}a}\left[\frac{\partial x'^{\alpha}}{\partial x^{\mu}}\right]\bigg|_{a=0} \tag{D.16}$$

$$\frac{\mathrm{d}}{\mathrm{d}a}\left[\mathrm{Det}\left[\frac{\partial x'}{\partial x}\right]\right]\bigg|_{a=0} = \frac{\mathrm{d}}{\mathrm{d}a}\left[\frac{\partial x'^{\alpha}}{\partial x^{\alpha}}\right]\bigg|_{a=0} = \frac{\mathrm{d}}{\mathrm{d}x^{\alpha}}\frac{\mathrm{d}x'^{\alpha}}{\mathrm{d}a}\bigg|_{a=0} \tag{D.17}$$

Thus,

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}a}S'_a[\phi']\bigg|_{a=0} &= \int_{\Omega}\mathrm{d}^4x\,\partial_{\mu}\left[\frac{\partial\mathcal{L}}{\partial\partial_{\mu}\phi(x)}\frac{\mathrm{d}\phi'(x)}{\mathrm{d}a}\bigg|_{a=0}\right] \\ &\quad + \int_{\Omega}\mathrm{d}^4x\,\mathcal{L}\frac{\mathrm{d}}{\mathrm{d}x^{\alpha}}\frac{\mathrm{d}x'^{\alpha}}{\mathrm{d}a}\bigg|_{a=0} \\ &\quad + \int_{\Omega}\mathrm{d}^4x\,\frac{\mathrm{d}\mathcal{L}}{\mathrm{d}x^{\alpha}}\frac{\mathrm{d}x'^{\alpha}}{\mathrm{d}a}\bigg|_{a=0}\end{aligned} \tag{D.18}$$

$$\frac{\mathrm{d}}{\mathrm{d}a}S'_a[\phi']\bigg|_{a=0} = \int_{\Omega}\mathrm{d}^4x\,\frac{\mathrm{d}}{\mathrm{d}x^{\mu}}\left[\frac{\partial\mathcal{L}}{\partial\partial_{\mu}\phi(x)}\frac{\mathrm{d}\phi'(x)}{\mathrm{d}a}\bigg|_{a=0} + \mathcal{L}\frac{\mathrm{d}x'^{\mu}}{\mathrm{d}a}\bigg|_{a=0}\right] \tag{D.19}$$

It's trivial to include indicies,

$$\frac{\mathrm{d}}{\mathrm{d}a^I}S'_a[\phi']\bigg|_{a^I=0} = \int_{\Omega}\mathrm{d}^4x\,\frac{\mathrm{d}}{\mathrm{d}x^{\mu}}\left[\frac{\partial\mathcal{L}}{\partial\partial_{\mu}\phi(x)}\frac{\mathrm{d}\phi'(x)}{\mathrm{d}a^I}\bigg|_{a^I=0} + \mathcal{L}\frac{\mathrm{d}x'^{\mu}}{\mathrm{d}a^I}\bigg|_{a^I=0}\right] \tag{D.20}$$

So that the conserved current is,

$$J^{\mu}{}_{I} = \frac{\partial\mathcal{L}}{\partial\partial_{\mu}\phi(x)}\frac{\mathrm{d}\phi'(x)}{\mathrm{d}a^I}\bigg|_{a^I=0} + \mathcal{L}\frac{\mathrm{d}x'^{\mu}}{\mathrm{d}a^I}\bigg|_{a^I=0} \tag{D.21}$$

E Noether's Second Theorem

This is an extension of the last result which deals with local transformations that leave the action invariant. Notice, we aren't saying symmetry, but invariance, this will be clear in the end of this appendix. Let the transformation be parametrized by a function $a(x)$,

$$\begin{cases} x^\mu & \to x'^\mu = x'^\mu(x; a(x)) \\ \phi(x) & \to \phi'(x') = \phi'(\phi(x), x; a(x)) \end{cases} \Rightarrow S[\phi] = S'[\phi'; a] \tag{E.1}$$

The reasoning proceeds as before, but now the derivatives are functional ones, also, due to distributional matters, is of need to include a test function,

$$\begin{aligned} \int_\Omega \mathrm{d}^4 z \, \frac{\delta S'[\phi'; a]}{\delta a(z)} \psi(z) = & \int_\Omega \mathrm{d}^4 z \, \psi(z) \int_\Omega \mathrm{d}^4 x \int_\Omega \mathrm{d}^4 y \, \frac{\delta \mathcal{L}}{\delta \phi(y)} \frac{\delta \phi'(y)}{\delta a(z)} \bigg|_{a \equiv 0} \\ & + \int_\Omega \mathrm{d}^4 z \, \psi(z) \int_\Omega \mathrm{d}^4 x \, \mathcal{L} \frac{\mathrm{d}}{\mathrm{d}x^\alpha} \frac{\delta x'^\alpha}{\delta a(z)} \bigg|_{a \equiv 0} \\ & + \int_\Omega \mathrm{d}^4 z \, \psi(z) \int_\Omega \mathrm{d}^4 x \, \frac{\mathrm{d}\mathcal{L}}{\mathrm{d}x^\alpha} \frac{\delta x'^\alpha}{\delta a(z)} \bigg|_{a \equiv 0} \end{aligned} \tag{E.2}$$

$$\begin{aligned} \int_\Omega \mathrm{d}^4 z \, \frac{\delta S'[\phi'; a]}{\delta a(z)} \psi(z) = & \int_\Omega \mathrm{d}^4 z \, \psi(z) \int_\Omega \mathrm{d}^4 x \int_\Omega \mathrm{d}^4 y \left[\frac{\partial \mathcal{L}}{\partial \phi(x)} \delta^{(4)}(x-y) + \frac{\partial \mathcal{L}}{\partial \partial_{x\mu} \phi(x)} \partial_{x\mu} \delta^{(4)}(x-y) \right] \frac{\delta \phi'(y)}{\delta a(z)} \bigg|_{a \equiv 0} \\ & + \int_\Omega \mathrm{d}^4 z \, \psi(z) \int_\Omega \mathrm{d}^4 x \, \frac{\mathrm{d}}{\mathrm{d}x^\alpha} \left[\mathcal{L} \frac{\delta x'^\alpha}{\delta a(z)} \bigg|_{a \equiv 0} \right] \end{aligned} \tag{E.3}$$

$$\begin{aligned} \int_\Omega \mathrm{d}^4 z \, \frac{\delta S'[\phi'; a]}{\delta a(z)} \psi(z) = & \int_\Omega \mathrm{d}^4 z \, \psi(z) \int_\Omega \mathrm{d}^4 x \int_\Omega \mathrm{d}^4 y \, \frac{\partial \mathcal{L}}{\partial \phi(x)} \delta^{(4)}(x-y) \frac{\delta \phi'(y)}{\delta a(z)} \bigg|_{a \equiv 0} \\ & + \int_\Omega \mathrm{d}^4 z \, \psi(z) \int_\Omega \mathrm{d}^4 x \int_\Omega \mathrm{d}^4 y \, \frac{\partial \mathcal{L}}{\partial \partial_{x\mu} \phi(x)} \partial_{x\mu} \delta^{(4)}(x-y) \frac{\delta \phi'(y)}{\delta a(z)} \bigg|_{a \equiv 0} \\ & + \int_\Omega \mathrm{d}^4 z \, \psi(z) \int_\Omega \mathrm{d}^4 x \, \frac{\mathrm{d}}{\mathrm{d}x^\alpha} \left[\mathcal{L} \frac{\delta x'^\alpha}{\delta a(z)} \bigg|_{a \equiv 0} \right] \end{aligned} \tag{E.4}$$

$$\begin{aligned} \int_\Omega \mathrm{d}^4 z \, \frac{\delta S'[\phi'; a]}{\delta a(z)} \psi(z) = & \int_\Omega \mathrm{d}^4 z \, \psi(z) \int_\Omega \mathrm{d}^4 x \, \frac{\partial \mathcal{L}}{\partial \phi(x)} \frac{\delta \phi'(x)}{\delta a(z)} \bigg|_{a \equiv 0} \\ & + \int_\Omega \mathrm{d}^4 z \, \psi(z) \int_\Omega \mathrm{d}^4 x \int_\Omega \mathrm{d}^4 y \, \partial_{x\mu} \left[\frac{\partial \mathcal{L}}{\partial \partial_\mu \phi(x)} \delta^{(4)}(x-y) \right] \frac{\delta \phi'(y)}{\delta a(z)} \bigg|_{a \equiv 0} \\ & - \int_\Omega \mathrm{d}^4 z \, \psi(z) \int_\Omega \mathrm{d}^4 x \int_\Omega \mathrm{d}^4 y \, \partial_{x\mu} \frac{\partial \mathcal{L}}{\partial \partial_\mu \phi(x)} \delta^{(4)}(x-y) \frac{\delta \phi'(y)}{\delta a(z)} \bigg|_{a \equiv 0} \\ & + \int_\Omega \mathrm{d}^4 z \, \psi(z) \int_\Omega \mathrm{d}^4 x \, \frac{\mathrm{d}}{\mathrm{d}x^\alpha} \left[\mathcal{L} \frac{\delta x'^\alpha}{\delta a(z)} \bigg|_{a \equiv 0} \right] \end{aligned} \tag{E.5}$$

$$\begin{aligned} \int_\Omega \mathrm{d}^4 z \, \frac{\delta S'[\phi'; a]}{\delta a(z)} \psi(z) = & \int_\Omega \mathrm{d}^4 z \, \psi(z) \int_\Omega \mathrm{d}^4 x \, E[\phi(x); x] \frac{\delta \phi'(x)}{\delta a(z)} \bigg|_{a \equiv 0} \\ & + \int_\Omega \mathrm{d}^4 z \, \psi(z) \int_\Omega \mathrm{d}^4 x \, \partial_{x\mu} \left[\frac{\partial \mathcal{L}}{\partial \partial_\mu \phi(x)} \frac{\delta \phi'(x)}{\delta a(z)} \bigg|_{a \equiv 0} \right] \end{aligned}$$

$$+\int_\Omega \mathrm{d}^4z\,\psi(z)\int_\Omega \mathrm{d}^4x\,\frac{\mathrm{d}}{\mathrm{d}x^\alpha}\left[\mathcal{L}\frac{\delta x'^\alpha}{\delta a(z)}\bigg|_{a\equiv 0}\right] \tag{E.6}$$

Usually the space-time isn't transformed in a local transformation,

$$\begin{aligned}0 = &\int_\Omega \mathrm{d}^4z\,\psi(z)\int_\Omega \mathrm{d}^4x\,E[\phi(x);x]\frac{\delta\phi'(x)}{\delta a(z)}\bigg|_{a\equiv 0}\\ &+\int_\Omega \mathrm{d}^4z\,\psi(z)\int_{\partial\Omega} \mathrm{d}^3S_{x\mu}\,\frac{\partial\mathcal{L}}{\partial\partial_\mu\phi(x)}\frac{\delta\phi'(x)}{\delta a(z)}\bigg|_{a\equiv 0}\end{aligned} \tag{E.7}$$

As an example, let's use the electromagnetic case,

$$\begin{aligned}0 = &\int_\Omega \mathrm{d}^4z\,\psi(z)\int_\Omega \mathrm{d}^4x\,E^\nu\frac{\delta A'_\nu(x)}{\delta a(z)}\bigg|_{a\equiv 0}\\ &+\int_\Omega \mathrm{d}^4z\,\psi(z)\int_{\partial\Omega} \mathrm{d}^3S_{x\mu}\,\frac{\partial\mathcal{L}}{\partial\partial_\mu A_\nu(x)}\frac{\delta A'_\nu(x)}{\delta a(z)}\bigg|_{a\equiv 0}\end{aligned} \tag{E.8}$$

$$\begin{aligned}0 = &\int_\Omega \mathrm{d}^4z\,\psi(z)\int_\Omega \mathrm{d}^4x\,E^\nu\partial_{x\nu}\delta^{(4)}(x-z)\\ &+\int_\Omega \mathrm{d}^4z\,\psi(z)\int_{\partial\Omega} \mathrm{d}^3S_{x\mu}\,\frac{\partial\mathcal{L}}{\partial\partial_\mu A_\nu(x)}\partial_{x\nu}\delta^{(4)}(x-z)\end{aligned} \tag{E.9}$$

$$\begin{aligned}0 = &-\int_\Omega \mathrm{d}^4z\,\psi(z)\int_\Omega \mathrm{d}^4x\,\partial_{x\nu}E^\nu\delta^{(4)}(x-z)\\ &+\int_\Omega \mathrm{d}^4z\,\psi(z)\int_{\partial\Omega} \mathrm{d}^3S_{x\nu}\,E^\nu\delta^{(4)}(x-z)\\ &-\int_\Omega \mathrm{d}^4z\,\psi(z)\int_{\partial\Omega} \mathrm{d}^3S_{x\mu}\,\partial_{x\nu}\frac{\partial\mathcal{L}}{\partial\partial_\mu A_\nu(x)}\delta^{(4)}(x-z)\end{aligned} \tag{E.10}$$

$$0 = -\int_\Omega \mathrm{d}^4x\,\partial_\mu E^\mu\psi(x) + \int_{\partial\Omega} \mathrm{d}^3S_\mu\left[E^\mu - \partial_\nu\frac{\partial\mathcal{L}}{\partial\partial_\mu A_\nu(x)}\right]\psi(x) \tag{E.11}$$

This is all without imposing the equations of motion,

F Boundary Conditions

Let our spacetime be a four dimensional manifold Ω, with boundary $\partial\Omega = \Sigma_- \cup \Sigma_+ \cup \Gamma$. Our action principle is given by,

$$S[\phi] = \int\limits_{\Omega} \mathrm{d}^4x\, \mathcal{L}[\phi(x), x] + \int\limits_{\partial\Omega} \mathrm{d}^3S_{x\alpha}\, \ell^\alpha[\phi(x), x] \tag{F.1}$$

The variation of the action cannot have contributions from the bulk of Γ, and also is done with compact support outside of $\Sigma_- \cup \Sigma_+$, that is, on $\tilde{\Omega} = \Omega' \cup \Gamma'$,

$$\int\limits_{\tilde{\Omega}} \mathrm{d}^4y\, \frac{\delta S[\phi]}{\delta\phi(y)}\psi(y) = \int\limits_{\tilde{\Omega}} \mathrm{d}^4y \int\limits_{\Omega} \mathrm{d}^4x\, \frac{\delta\mathcal{L}[x]}{\delta\phi(y)}\psi(y) + \int\limits_{\tilde{\Omega}} \mathrm{d}^4y \int\limits_{\partial\Omega} \mathrm{d}^3S_{x\alpha}\, \frac{\delta\ell^\alpha[x]}{\delta\phi(y)}\psi(y) \tag{F.2}$$

$$\begin{aligned}\int\limits_{\tilde{\Omega}} \mathrm{d}^4y\, \frac{\delta S[\phi]}{\delta\phi(y)}\psi(y) = &\int\limits_{\tilde{\Omega}} \mathrm{d}^4y \int\limits_{\Omega} \mathrm{d}^4x \left[\frac{\partial\mathcal{L}[x]}{\partial\phi(x)}\delta^{(4)}(x-y) + \frac{\partial\mathcal{L}[x]}{\partial\partial_\mu\phi(x)}\partial_\mu\delta^{(4)}(x-y)\right]\psi(y) \\ &+ \int\limits_{\tilde{\Omega}} \mathrm{d}^4y \int\limits_{\partial\Omega} \mathrm{d}^3S_{x\alpha}\, \frac{\delta\ell^\alpha[x]}{\delta\phi(y)}\psi(y)\end{aligned} \tag{F.3}$$

$$\begin{aligned}\int\limits_{\tilde{\Omega}} \mathrm{d}^4y\, \frac{\delta S[\phi]}{\delta\phi(y)}\psi(y) = &\int\limits_{\tilde{\Omega}} \mathrm{d}^4y \int\limits_{\Omega} \mathrm{d}^4x \left[\frac{\partial\mathcal{L}[x]}{\partial\phi(x)} - \partial_\mu\frac{\partial\mathcal{L}[x]}{\partial\partial_\mu\phi(x)}\right]\delta^{(4)}(x-y)\psi(y) \\ &+ \int\limits_{\tilde{\Omega}} \mathrm{d}^4y \int\limits_{\Omega} \mathrm{d}^4x\, \partial_{x\mu}\left[\frac{\partial\mathcal{L}[x]}{\partial\partial_\mu\phi(x)}\delta^{(4)}(x-y)\right]\psi(y) + \int\limits_{\tilde{\Omega}} \mathrm{d}^4y \int\limits_{\partial\Omega} \mathrm{d}^3S_{x\alpha}\, \frac{\delta\ell^\alpha[x]}{\delta\phi(y)}\psi(y)\end{aligned} \tag{F.4}$$

$$\begin{aligned}\int\limits_{\tilde{\Omega}} \mathrm{d}^4y\, \frac{\delta S[\phi]}{\delta\phi(y)}\psi(y) = &\int\limits_{\tilde{\Omega}} \mathrm{d}^4y \left[\frac{\partial\mathcal{L}[y]}{\partial\phi(y)} - \partial_\mu\frac{\partial\mathcal{L}[y]}{\partial\partial_\mu\phi(y)}\right]\psi(y) \\ &+ \int\limits_{\tilde{\Omega}} \mathrm{d}^4y \int\limits_{\partial\Omega} \mathrm{d}^3S_{x\alpha}\, \frac{\partial\mathcal{L}[x]}{\partial\partial_\alpha\phi(x)}\delta^{(4)}(x-y)\psi(y) + \int\limits_{\tilde{\Omega}} \mathrm{d}^4y \int\limits_{\partial\Omega} \mathrm{d}^3S_{x\alpha}\, \frac{\delta\ell^\alpha[x]}{\delta\phi(y)}\psi(y)\end{aligned} \tag{F.5}$$

$$\begin{aligned}\int\limits_{\tilde{\Omega}} \mathrm{d}^4y\, \frac{\delta S[\phi]}{\delta\phi(y)}\psi(y) = &\int\limits_{\tilde{\Omega}} \mathrm{d}^4y \left[\frac{\partial\mathcal{L}[y]}{\partial\phi(y)} - \partial_\mu\frac{\partial\mathcal{L}[y]}{\partial\partial_\mu\phi(y)}\right]\psi(y) \\ &+ \int\limits_{\partial\Omega} \mathrm{d}^3S_{x\alpha}\, \frac{\partial\mathcal{L}[x]}{\partial\partial_\alpha\phi(x)}\psi(x) + \int\limits_{\tilde{\Omega}} \mathrm{d}^4y \int\limits_{\partial\Omega} \mathrm{d}^3S_{x\alpha}\, \frac{\delta\ell^\alpha[x]}{\delta\phi(y)}\psi(y)\end{aligned} \tag{F.6}$$

For many cases $\ell^\alpha \equiv 0$ due to dynamical and covariance reasons. Of course ψ is zero on $\Sigma_\pm$, then, the equations of motion impose:

$$0 = \frac{\partial\mathcal{L}[\phi(x); x]}{\partial\phi(x)} - \partial_\mu\frac{\partial\mathcal{L}[\phi(x); x]}{\partial\partial_\mu\phi(x)}\bigg|_{x\in\Omega} \tag{F.7}$$

$$0 = n_\alpha(x)\partial^\alpha\phi(x)\bigg|_{x\in\Gamma} \tag{F.8}$$

With n_α being the normal direction in Γ. If we try to compute the conserved charge by Noether's theorem,

$$0 = \int\limits_{\partial\Omega} \mathrm{d}^3 S_\mu \left[\frac{\partial \mathcal{L}}{\partial \partial_\mu \phi(x)} \frac{\mathrm{d}\phi'(x)}{\mathrm{d}a} \bigg|_{a=0} + \mathcal{L} \frac{\mathrm{d}x'^\mu}{\mathrm{d}a} \bigg|_{a=0} \right] \tag{F.9}$$

$$0 = \int\limits_{\Sigma_+} \mathrm{d}^3 S_\mu \, J^\mu - \int\limits_{\Sigma_-} \mathrm{d}^3 S_\mu \, J^\mu + \int\limits_{\Gamma} \mathrm{d}^3 S_\mu \left[\frac{\partial \mathcal{L}}{\partial \partial_\mu \phi(x)} \frac{\mathrm{d}\phi'(x)}{\mathrm{d}a} \bigg|_{a=0} + \mathcal{L} \frac{\mathrm{d}x'^\mu}{\mathrm{d}a} \bigg|_{a=0} \right] \tag{F.10}$$

$$0 = \int\limits_{\Sigma_+} \mathrm{d}^3 S_\mu \, J^\mu - \int\limits_{\Sigma_-} \mathrm{d}^3 S_\mu \, J^\mu + \int\limits_{\Gamma} \mathrm{d}^3 S_\mu \, \mathcal{L} \frac{\mathrm{d}x'^\mu}{\mathrm{d}a} \bigg|_{a=0} \tag{F.11}$$

Now, if we have $\mathcal{L}\big|_{x\in\Gamma} = 0$ or $\frac{\mathrm{d}x'}{\mathrm{d}a}\big|_{a=0,x\in\Gamma} = 0$ then we get a conserved charge!

$$\int\limits_{\Sigma_+} \mathrm{d}^3 S_\mu \, J^\mu = \int\limits_{\Sigma_-} \mathrm{d}^3 S_\mu \, J^\mu \tag{F.12}$$

Now let's consider something other, let be a variation with change the values of the fields at the time-like boundaries, that is, a variation between different solutions,

$$\begin{aligned} \int\limits_{\Omega} \mathrm{d}^4 y \, \frac{\delta S[\phi]}{\delta \phi(y)} \psi_1(y) = & \int\limits_{\Omega} \mathrm{d}^4 y \, E[\phi(y); y] \psi_1(y) \\ & + \int\limits_{\partial\Omega} \mathrm{d}^3 S_\alpha \, \Theta^\alpha[\phi(x); x] \psi_1(x) \end{aligned} \tag{F.13}$$

$$\begin{aligned} \int\limits_{\Omega} \mathrm{d}^4 y \int\limits_{\Omega} \mathrm{d}^4 z \, \frac{\delta^2 S[\phi]}{\delta\phi(z)\delta\phi(y)} \psi_1(y)\psi_2(z) = & \int\limits_{\Omega} \mathrm{d}^4 y \int\limits_{\Omega} \mathrm{d}^4 z \, \frac{\delta E[\phi(y); y]}{\delta\phi(z)} \psi_1(y)\psi_2(z) \\ & + \int\limits_{\partial\Omega} \mathrm{d}^3 S_\alpha \int\limits_{\Omega} \mathrm{d}^4 z \, \frac{\delta \Theta^\alpha[\phi(x); x]}{\delta\phi(z)} \psi_1(x)\psi_2(z) \end{aligned} \tag{F.14}$$

$$\tag{F.15}$$

Antisymmetrize in $1 \leftrightarrow 2$,

$$\begin{aligned} 0 = & \int\limits_{\Omega} \mathrm{d}^4 y \int\limits_{\Omega} \mathrm{d}^4 z \left(\frac{\delta E[\phi(y); y]}{\delta\phi(z)} - \frac{\delta E[\phi(z); z]}{\delta\phi(y)} \right) \psi_1(y)\psi_2(z) \\ & + \int\limits_{\partial\Omega} \mathrm{d}^3 S_\alpha \int\limits_{\Omega} \mathrm{d}^4 z \left(\frac{\delta \Theta^\alpha[\phi(x); x]}{\delta\phi(z)} - \frac{\delta \Theta^\alpha[\phi(z); z]}{\delta\phi(x)} \right) \psi_1(x)\psi_2(z) \end{aligned} \tag{F.16}$$

As long as the equation of motion have just even order of derivatives with constant coefficients,

$$0 = \int\limits_{\partial\Omega} \mathrm{d}^3 S_\alpha \int\limits_{\Omega} \mathrm{d}^4 z \left(\partial_x^\alpha \delta^{(4)}(x-z) - \partial_z^\alpha \delta^{(4)}(z-x) \right) \psi_1(x)\psi_2(z) \tag{F.17}$$

$$0 = \int\limits_{\partial\Omega} \mathrm{d}^3 S_\alpha \left(\psi_1(x) \partial^\alpha \psi_2(x) - \psi_2(x) \partial^\alpha \psi_1(x) \right) \tag{F.18}$$

Suppose that for a given solution of the equations of motion ϕ, it's true that $\phi + \psi_i$ are also solutions of the equations of motion, then,

$$\int_{\Sigma_+} \mathrm{d}^3 S_\alpha \left(\psi_1(x)\partial^\alpha \psi_2(x) - \psi_2(x)\partial^\alpha \psi_1(x)\right) = \int_{\Sigma_-} \mathrm{d}^3 S_\alpha \left(\psi_1(x)\partial^\alpha \psi_2(x) - \psi_2(x)\partial^\alpha \psi_1(x)\right) \tag{F.19}$$

In other words, the functional,

$$\Omega(\psi_1, \psi_2) = \int_{\Sigma_t} \mathrm{d}^3\mathbf{x} \left(\psi_1 \dot{\psi}_2 - \psi_2 \dot{\psi}_1\right) \tag{F.20}$$

defined in the tangent space of the phase space of solutions, is independent of the time slice used to compute it. Taken into account that the phase space needs two variables for each degree of freedom, they are chosen as $\phi(\mathbf{x})$ and $\pi(\mathbf{x}) = \dot{\phi}(\mathbf{x})$, in this way, our functional can be rewritten as a phase space 2-form,

$$\Omega = \int_{\Sigma_t} \mathrm{d}^3\mathbf{x}\, \delta\phi(\mathbf{x}) \wedge \delta\pi(\mathbf{x}) \tag{F.21}$$

The main point is that this 2-form is non degenerated and can be inverted as,

$$\{\cdot, \cdot\} = \int_{\Sigma_t} \mathrm{d}^3\mathbf{x} \left(\frac{\delta}{\delta\phi(\mathbf{x})}\frac{\delta}{\delta\pi(\mathbf{x})} - \frac{\delta}{\delta\pi(\mathbf{x})}\frac{\delta}{\delta\phi(\mathbf{x})}\right) \tag{F.22}$$

Which provides the fundamental Poisson bracket,

$$\{\phi(t, \mathbf{x}), \pi(t, \mathbf{y})\} = \delta^{(3)}(\mathbf{x} - \mathbf{y}) \tag{F.23}$$

G Algebra of conserved quantities

Given an internal symmetry of our action,

$$\begin{cases} \phi^I(x) & \to \phi'^I(x;a) \\ x^\mu & \to x'^\mu = x^\mu \end{cases} \Rightarrow S[\phi] = S'_a[\phi] \tag{G.1}$$

where I encode whatever index the fields of our theory may have. Suppose this symmetry is represented by some Lie Group/Algebra, that is, we have,

$$\phi'^I(x) = \exp\left(-\mathrm{i}a_i T^i\right)^I{}_J \phi^J(x) \Rightarrow \left.\frac{\mathrm{d}\phi'^I(x)}{\mathrm{d}a^i}\right|_{a^i=0} = -\mathrm{i}T^I{}_{Ji}\,\phi^J(x) \tag{G.2}$$

Consider the conserved current associated with this symmetry,

$$J^\mu{}_i = \frac{\partial\mathcal{L}}{\partial\partial_\mu\phi^I(x)}\left.\frac{\mathrm{d}\phi'^I(x)}{\mathrm{d}a^i}\right|_{a^i=0} \tag{G.3}$$

$$J^\mu{}_i = (-)^2\partial^\mu\phi_I(x)\mathrm{i}T^I{}_{Ji}\,\phi^J(x) \tag{G.4}$$

$$J^0{}_i = -\pi_I\mathrm{i}T^I{}_{Ji}\,\phi^J \tag{G.5}$$

Where we made use of the canonical conjugated momenta $\pi_I = \frac{\partial\mathcal{L}}{\partial\dot{\phi}^I}$. Now compute the conserved charge,

$$Q_i = \int \mathrm{d}^3\mathbf{x}\, J^0{}_i = -\int \mathrm{d}^3\mathbf{x}\,\pi_I\mathrm{i}T^I{}_{Ji}\,\phi^J \tag{G.6}$$

Remember from classical mechanics, in the Hamiltonian formalism we have as dynamical variables the generalized coordinate $\phi^I(t,\mathbf{x})$, and it's canonical conjugated momentum $\pi_I(t,\mathbf{x})$, they define the phase space and can be used to construct the Poisson bracket,

$$\{A,B\} = \int \mathrm{d}^3\mathbf{x}\left(\frac{\delta A}{\delta\phi^I(\mathbf{x})}\frac{\delta B}{\delta\pi_I(\mathbf{x})} - \frac{\delta A}{\delta\pi_I(\mathbf{x})}\frac{\delta A}{\delta\phi^I(\mathbf{x})}\right) \tag{G.7}$$

as this is a theory of the continuum, derivatives have to be exchanged by functional derivatives and sum over canonical pairs have to be exchanged for integration, still, it satisfy the canonical equal time Poisson bracket relations, $\left\{\phi^I(t,\mathbf{x}),\pi_J(t,\mathbf{y})\right\} = \delta^I{}_J\delta^{(3)}(\mathbf{x}-\mathbf{y})$ (verify!). It's interesting to compute the Poisson bracket between two conserved charges,

$$\{Q_i,Q_j\} = \int \mathrm{d}^3\mathbf{x}\left(\frac{\delta Q_i}{\delta\phi^I(\mathbf{x})}\frac{\delta Q_j}{\delta\pi_I(\mathbf{x})} - \frac{\delta Q_i}{\delta\pi_I(\mathbf{x})}\frac{\delta Q_j}{\delta\phi^I(\mathbf{x})}\right) \tag{G.8}$$

$$\{Q_i,Q_j\} = (-)^2\int \mathrm{d}^3\mathbf{x}\left(\pi_K\mathrm{i}T^K{}_{Ii}\,\mathrm{i}T^I{}_{Jj}\,\phi^J - \mathrm{i}T^I{}_{Ji}\,\phi^J\pi_K\mathrm{i}T^K{}_{Ij}\right) \tag{G.9}$$

$$\{Q_i,Q_j\} = \mathrm{i}^2\int \mathrm{d}^3\mathbf{x}\,\pi_K\left(T^K{}_{Ii}T^I{}_{Jj} - T^K{}_{Ij}T^I{}_{Ji}\right)\phi^J \tag{G.10}$$

$$\{Q_i,Q_j\} = \mathrm{i}^2\int \mathrm{d}^3\mathbf{x}\,\pi_K\Big([T_i,T_j]^K{}_J\Big)\phi^J \tag{G.11}$$

Now, as $T^K{}_{Ii}$ are the generators of a Lie Algebra, they obey an commutation relation of the form, $[T_i,T_j] = \mathrm{i}f_{ij}{}^kT_k$, thus,

$$\{Q_i,Q_j\} = \mathrm{i}^2\int \mathrm{d}^3\mathbf{x}\,\pi_K\left(\mathrm{i}f_{ij}{}^kT^K{}_{Jk}\right)\phi^J \tag{G.12}$$

$$\{Q_i, Q_j\} = -\mathrm{i}^2 f_{ij}{}^k (-) \int \mathrm{d}^3\mathbf{x}\, \pi_K \mathrm{i} T^K{}_{Jk}\, \phi^J \tag{G.13}$$

$$\{Q_i, Q_j\} = f_{ij}{}^k Q_k \tag{G.14}$$

The classical charges satisfy an algebra with the same structure constants as the symmetry group! In the quantum theory, at least from the canonical quantization POV, we have a correspondence between the commutator of two operators, and the Poisson bracket between the classical observables, $\left[\hat{A}, \hat{B}\right] \to \mathrm{i}\hbar\{A, B\}$, therefore, each classical conserved charge should be correspondent to a conserved charge operator $\hat{Q}_i$ which *should* satisfy the same Lie Algebra,

$$\left[\hat{Q}_i, \hat{Q}_j\right] = \mathrm{i}\hbar f_{ij}{}^k \hat{Q}_k \tag{G.15}$$

Almost always this isn't the case, and the symmetry has a anomaly in the quantum theory, that is, a transformation which is a symmetry in the classical theory isn't anymore a symmetry when the theory is quantized. For these cases, the most common and physical relevant way an anomaly shows is through a central extension of the Lie Algebra,

$$\left[\hat{Q}_i, \hat{Q}_j\right] = \mathrm{i}\hbar f_{ij}{}^k \hat{Q}_k + \hbar^2 C_{[ij]} \mathbb{1} \tag{G.16}$$

The name *central extension* is so because $C_{[ij]}$ are just numbers, that is, they commute with every other generator of the algebra, and thus, they belong to the center of the algebra — the maximum set of mutually commuting generators —. The $\hbar^2$ is needed for dimensional reasons (check it!), and it explain why this don't show up in the classical theory, if we consider the limit of,

$$\{Q_i, Q_f\} := \lim_{\hbar \to 0} \frac{1}{\mathrm{i}\hbar} \left[\hat{Q}_i, \hat{Q}_j\right] = f_{ij}{}^k Q_k \tag{G.17}$$

That is, the $\hbar^2$ term drop off, it's a true quantum effect. There are other ways of understanding why the central extensions of algebras show up in the process of quantization, there is a close relation with the need to consider the double cover of Lie Groups/Algebras when going from classical symmetries to quantum symmetries, and also some overlap with the need of normal ordering quantum operators. Most of the anomalies are harmless, but, if there is an gauge anomaly the theory is physically inconsistent.

Luckily, the Poincaré Group/Algebra does not allow any kind of central extension (see Weinberg for proof), hence, the Poincaré symmetry is realized correctly in the quantum theory.

Other relevant property of the conserved charges is: they generate the symmetry transformation in the phase space variables,

$$\left\{Q_i, \phi^I(x)\right\} = \int \mathrm{d}^3\mathbf{y} \left(\frac{\delta Q_i}{\delta \phi^K(\mathbf{y})} \frac{\delta \phi^I(t, \mathbf{x})}{\delta \pi_K(\mathbf{y})} - \frac{\delta Q_i}{\delta \pi_K(\mathbf{y})} \frac{\delta \phi^I(t, \mathbf{x})}{\delta \phi^K(\mathbf{y})} \right) \tag{G.18}$$

$$\left\{Q_i, \phi^I(x)\right\} = (-)^2 \int \mathrm{d}^3\mathbf{y}\, \mathrm{i} T^K{}_{Ji}\, \phi^J \delta^I{}_K \delta^{(3)}(\mathbf{x} - \mathbf{y}) \tag{G.19}$$

$$\left\{Q_i, \phi^I(x)\right\} = \mathrm{i} T^I{}_{Ji}\, \phi^J \equiv \left.\frac{\mathrm{d}\phi'^I}{\mathrm{d}a^i}\right|_{a^I=0} \Rightarrow \left[\hat{Q}_i, \hat{\phi}^I(x)\right] = \mathrm{i}\hbar \left.\frac{\mathrm{d}\hat{\phi}'^I}{\mathrm{d}a^i}\right|_{a^I=0} \tag{G.20}$$

The same reasoning can be applied to space-time symmetries (try it!).